TACT: Trust-Anchored Confidence Tempering for Self-Consistency Voting in Large Language Models

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Abstract—Confidence-weighted self-consistency (CISC and its successors) improves on majority voting when a frozen large language model's self-reported confidence is calibrated in direction. Every published weighting scheme is structurally monotone increasing in confidence, so an anti-correlated channel poisons the vote instead of informing it, while binary dev-set gates survive inversion only by discarding genuinely discriminative signal. This paper presents tact (Trust-Anchored Confidence Tempering), which replaces the fixed confidence exponent with one derived from the measured, signed, within-item discrimination of the channel: a pooled van Elteren Somers' D rank statistic with an item-clustered standard error, passed through positive-part James–Stein shrinkage and a Bayes-discriminant link. Written out, the method is a single expression whose exponent reduces to $\gamma = z\sqrt{2 + z^2}$ with z the probit of the shrunk pooled AUC, and it carries exact anchors: inside the shrinkage dead zone the vote is bit-identical to plain self-consistency, and a log-value feature map reproduces CISC-power. A label-free variant estimates the signed reliability from agreement pseudo-labels under a proven attenuation identity that guarantees sign consistency whenever the plurality-error rate is below one half, with conservative de-attenuation and echo alarms at the identifiability boundary. On a synthetic-oracle harness with paired trace pools, the label-free variant recovers anti-correlated channels that pin every published protocol to the majority-vote floor ($\kappa = -0.6$: 1.000 vs. 0.807), rank invariance beats the best exponent in the tested raw-value grid $\{0.25, 0.5, 1, 2, 4\}$ under monotone confidence compression (1.000 vs. 0.965, the grid optimum sitting at its upper endpoint), and a per-group extension cracks the heterogeneity floor with zero paired losses to self-consistency (0.940 vs. 0.808; $+79/-0$, $p = 3.3 \times 10^{-24}$). Two real-trace campaigns on a frozen model confirm the premise and locate the binding constraint: within-item discrimination is positive on competition mathematics (pooled $\widehat{D} = +0.250$, $z = +2.54$), yet the stratum on which any such method can act, where the plurality is wrong and the correct answer is present in the pool, measures 3–7.5% of items across five substrates in two domains, code generation with executable ground truth included. Abstention is therefore the correct default rather than a conservative one, and the dead zone implements it exactly. The paper further proves that per-item label-free adaptation is impossible under i.i.d. latent coupling, and pre-registers four falsification criteria, among them the published dev-calibrated CISC protocol as a designated killer baseline, all of which the method survived.

Index Terms—large language models, self-consistency, confidence calibration, weighted voting, label-free estimation, rank statistics

I. Introduction

Self-consistency (sc) [1] improves the reasoning accuracy of a frozen large language model (LLM) by sampling K chain-of-thought traces and returning the plurality answer. Because each trace can also report a confidence score (verbalized [2], [3], derived from token log-probabilities, or elicited as $P(\text{True})$ [4]), a natural refinement is to weight votes by confidence. Confidence-Informed Self-Consistency (CISC) [5] showed that this recovers the accuracy of plain sc at a fraction of the sampling budget, and introduced Within-Question Discrimination (WQD) to argue that discrimination, not calibration, is the property that makes a confidence signal useful for voting.

This refinement carries a structural fragility that, to the author's knowledge, no published method addresses. Every existing weighting scheme is monotone increasing in confidence, including CISC's softmax weights, reliability-aware pseudo-counts [6], and warmup-thresholded filtering [7]. The trust decision is which magnitude of up-weighting to apply; the possibility that the channel is anti-correlated with correctness is not representable. Yet miscalibration of direction is not exotic: reinforcement fine-tuning is known to distort verbalized confidence, distribution shift can invert a signal that was informative in-domain, and in the experiments reported here a simple anti-correlated channel ($\kappa = -0.6$; Section III) drives confidence-weighted baselines from near-perfect accuracy to far below the majority-vote floor, while the same evidence, read with the correct sign, is a perfect signal. The defensive alternative, a binary dev-set gate that disables the channel when calibration error is high, survives the inversion but discards discriminative signal wholesale: a systematically under-confident yet perfectly ranked channel fails an ECE gate for reasons irrelevant to voting utility [5], [8].

This paper frames the problem as estimating one scalar: the signed within-item discrimination of the confidence channel, and mapping that scalar, with its uncertainty, to a vote exponent. The contributions are:

C1: Signed, analytically-tempered confidence weighting. tact votes with weights $w_i = \exp(\gamma\,\varphi_i)$, where φ_i is the standardized van der Waerden score of trace i's within-item confidence midrank, and γ is derived, not grid-searched: a pooled van Elteren Somers' D statistic (equal to $2 \cdot \text{WQD} - 1$) with an exact tie-corrected null variance and an item-clustered jackknife standard error, shrunk by positive-part James–Stein with a significance floor, then mapped through a Bayes-discriminant link with a mixture-variance correction. The construction carries exact anchors: inside the shrinkage dead zone the vote is bit-identical to plain sc (a shared code path), and the log-value feature map reproduces CISC-power exactly (Section IV). Because φ depends on confidence only through within-item ranks, the entire method is invariant to every strictly monotone distortion of the confidence scale; under monotone compression it beats the best exponent in the tested raw-value grid $\{0.25, 0.5, 1, 2, 4\}$ (1.000 vs. 0.965); that grid optimum lies at the largest exponent tried, so the comparison bounds the grid, not the whole family.

C2: Label-free estimation of the signed reliability. The crowdsourcing lineage estimates annotator reliability from cross-annotator covariance [9], [10]; a single exchangeable confidence channel from one model offers no such structure. The signed discrimination is estimated from agreement pseudo-labels (deduplication-weighted plurality per item) with a proven class-conditional-noise attenuation identity, $\mathbb{E}[\widehat{D}_g] = (1 - 2\bar{\rho})\, D$: the label-free estimate can only under-trust, never mis-sign, whenever the pair-weighted plurality-error rate $\bar{\rho}$ is below 1/2. A split-half agreement inversion de-attenuates conservatively, and sign-aware alarms return the method to plain sc when identifiability is threatened. On the coupling sweep the label-free variant matches the 200-label variant nearly point-for-point, including full recovery of negative channels (Section VIII).

C3: An impossibility result and its structured escape. When the per-item coupling is i.i.d. with no observable covariate, per-item label-free adaptation is shown to be closed: any monotone use of an item's own agreement statistic collapses to plurality reinforcement; on exactly the plurality-wrong items where a flip could help, the observable sign opposes the truth 96% of the time; and the two hypotheses $\{\kappa > 0, \text{minority correct}\}$ and $\{\kappa < 0, \text{plurality correct}\}$ induce identical observable laws. When heterogeneity is instead indexed by an observable covariate (domain-dependent calibration), running the same estimator per group recovers each group's signed coupling and approaches the per-item oracle with zero paired losses to sc (Section VI).

C4: A pre-registered falsification protocol. Four falsifiers were fixed before implementation, including the two designed to kill the method: the published dev-calibrated CISC protocol (whose tuned temperature already interpolates sc↔CISC) and a trivial dev-picked signed exponent grid. All four survived, and the honest margins are reported: against the signed grid the net advantage concentrates in three cells: monotone distortion, confident echo, and label-free operation, which no grid can perform.

II. Related Work

Confidence-weighted self-consistency. sc [1] treats sampled traces as i.i.d. votes. CISC [5] weights votes by softmax-normalized confidence with a temperature tuned on a labeled split, and its WQD metric makes the discrimination-vs-calibration point that also motivates this work; the rank-calibration line [8] reaches the same conclusion independently. Weighted variants [11], [12] and early-stopping families [13], [14] refine the budget; reliability-aware pseudo-counts [6] and warmup-thresholded filtering [7] adapt online but only re-scale positive trust. None of these can represent, much less estimate, a negative confidence–correctness association. The dev-calibrated variant must therefore be positioned honestly: CISC's tuned temperature is already a dev-calibrated sc↔CISC interpolation, so the novelty of tact-dev lies in the sign, the rank invariance, and the analytic (grid-free) map, not in dev calibration itself.

Reliability estimation without labels. Estimating worker reliability from agreement is classical [9], [15], [16]; spectral meta-learners [10] and recent LLM ensemble work [17], [18] exploit covariance across multiple predictors. The setting here differs: one exchangeable channel from one model, per-item vote structure, and the known failure of agreement proxies under correlated errors—met here with a quantified attenuation identity, conservative de-attenuation, and alarms in place of an unconditional claim.

Shrinkage and rank statistics. The estimator assembles classical parts: stratified rank statistics [19], the James–Stein positive-part estimator [20], effective-sample-size corrections [21], [22], and normal-scores discriminant analysis. The claim is the assembly and its anchors, not the parts.

Honest sibling result. A preceding system by the author (RLEV-VoI, redundancy-discounted voting with value-of-information stopping) was evaluated under the same falsification discipline and failed it, dominated everywhere by a simple deduplication baseline, and is reported as a negative result. Its post-mortem isolated the confidence dilemma studied here.

III. Problem Setup

A. Notation

Items $q = 1, \ldots, Q$; item q has m_q sampled traces. Trace (q, i) yields an answer $a_{q,i}$ in a discrete set and a confidence $c_{q,i} \in (0, 1)$; correctness is $y_{q,i} = \mathbf{1}[a_{q,i} = a_q^*]$, unobserved at test time. Plain sc returns $\arg\max_A n_q(A)$ where $n_q(A)$ counts votes for answer A. CISC-power weights votes by $c_{q,i}^{\gamma}$ with a fixed $\gamma > 0$.

B. The confidence dilemma

The synthetic oracle draws, per item, traces from a cluster mixture with a latent correct answer and generates confidence as

$$c_{q,i} = \mathrm{clip}\big(\tfrac{1}{2} + \kappa\,(y_{q,i} - \tfrac{1}{2}) + \varepsilon_{q,i},\, 0.01,\, 0.99\big), \quad (1)$$

with noise $\varepsilon \sim \mathcal{N}(0, 0.1^2)$ and coupling $\kappa \in [-0.6, 0.6]$. Fig. 1 maps the baseline landscape before the proposed method existed: unconditional weighting (CISC, $\gamma = 1$) collapses on $\kappa < 0$; an ECE gate never opens off the well-calibrated diagonal; a sign-corrected AUC gate over dev labels nearly saturates the homogeneous sweep. This pre-measurement fixes where a new method can legitimately

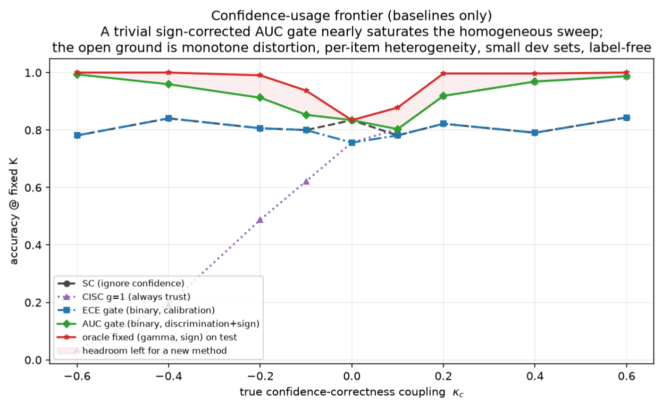

Fig. 1. The pre-measured problem statement: accuracy of baseline confidence policies at fixed K=15 as the true coupling κ varies. A trivial sign-corrected AUC gate (green) nearly saturates the homogeneous sweep; the headroom for any new method (shaded) concentrates in the mid-range and, off this plot, in distortion, heterogeneity, and label-free cells.

claim wins—monotone distortion of the confidence scale, covariate heterogeneity, small dev sets, and label-free operation—and the evaluation holds itself to exactly those cells.

IV. TACT

A. Vote family

Within item q, let $R_{q,i}$ be the midrank of $c_{q,i}$ (ties averaged) and

$$\varphi_{q,i} = \frac{v_{q,i} - \bar v_q}{\sigma_q}, \qquad v_{q,i} = \Phi^{-1}\Big(\frac{R_{q,i}}{m_q+1}\Big), \tag{2}$$

where σ_q is the realized standard deviation of v within the item (the no-tie value is 0.62 at m=4 but 0.95 at m=40; a closed form would silently rescale γ across budgets), and $\varphi \equiv 0$ if $\sigma_q \le 10^{-8}$ (all-tied confidences vote as plain sc). The vote is

$$\hat a_q = \arg\max_A \sum_{i:\, a_{q,i}=A} \exp\big(\gamma\,\varphi_{q,i}\big), \tag{3}$$

and when $\gamma = 0$ the implementation calls the sc routine itself, making the zero-trust anchor bitwise exact rather than equal in distribution. Because (2) depends on c only through within-item ranks, every strictly monotone distortion of the confidence scale leaves (3) unchanged.

B. Reliability statistic

For item q with n_q^1 positive and n_q^0 negative labels (dev: y; label-free: the pseudo-label of Section V), the Mann–Whitney statistic on midranks gives

$$D_q = 2\,\mathrm{AUC}_q - 1, \qquad \mathrm{AUC}_q = \frac{U_q}{n_q^1 n_q^0}, \tag{4}$$

which equals $2 \cdot \mathrm{WQD}_q - 1$ in CISC's notation. Pooling uses van Elteren pair-count weights $N_q = n_q^1 n_q^0$ [19]:

$$\widehat D = \frac{\sum_q N_q D_q}{\sum_q N_q}. \tag{5}$$

Under the within-item exchangeability null, U_q has the exact tie-corrected variance $n_q^1 n_q^0 (m_q{+}1)/12 \cdot [1 - \sum_t (t^3 - t)/(m_q^3 {-} m_q)]$, yielding a null standard error SE_0; between-item heterogeneity is captured by the closed-form delete-one-item jackknife SE_J. The conservative choice is

$$\mathrm{SE} = \max\big(\mathrm{SE}_0,\ \mathrm{SE}_J,\ \tfrac{1}{2\sqrt{N}}\big), \qquad r = \widehat D/\mathrm{SE}. \tag{6}$$

Because D is a pairwise functional, $\mathbb{E}[\widehat D]$ does not depend on m_q: an exponent estimated at m=40 transfers to deployment at m=8.

C. Tempering map

Shrinkage. Positive-part James–Stein with a significance floor ν:

$$\tilde D = \mathrm{sign}(\widehat D)\,\max\big(0,\ |\widehat D| - \nu^2 \mathrm{SE}^2/|\widehat D|\big), \tag{7}$$

with dead zone $\{|r| \le \nu\}$; $\nu_{\mathrm{dev}} = 1.28$, $\nu_{\mathrm{LF}} = 2.33$. With $\nu = 1$, (7) is exactly the empirical-Bayes posterior mean under a $\mathcal{N}(0, \tau^2)$ prior with plug-in $\hat\tau^2 = \max(0, \widehat D^2 - \mathrm{SE}^2)$ [20]. The map is odd, continuous, never exceeds $|\widehat D|$, and is monotone in $\widehat D$ and anti-monotone in SE.

Link. Model $\varphi \,|\, y \sim \mathcal{N}(\mu_y, s^2)$ within item with the mixture standardized to unit variance, which is what (2) enforces, so $s^2 = 1/(1 + \bar p(1-\bar p)u^2)$ where $u = \sqrt{2}\,\Phi^{-1}\big(\frac{1+\tilde D}{2}\big)$ and $\bar p$ is the base rate of correct traces. The Bayes-optimal per-trace log-weight coefficient is then

$$\gamma^* = \frac{u}{s} = u\sqrt{1 + \bar p(1-\bar p)\,u^2}, \tag{8}$$

capped at $\gamma_{\max}$ (4 dev, 2 label-free). The uncorrected link $\gamma = u$ under-weights strong channels by up to $\sim$50% at $D = 0.9$.

D. tact in one expression

Two simplifications collapse the pipeline. Factoring $\widehat D$ out of (7) makes the shrinkage a multiplicative gain in the pooled z-statistic $\zeta = \widehat D/\mathrm{SE}$ alone, and substituting $u = \sqrt{2}\,z$ with $z = \Phi^{-1}\big(\frac{1+\tilde D}{2}\big)$ into (8) removes the nested radical. tact is then

$$\boxed{\hat a_q = \arg\max_A \sum_{i:\, a_{q,i}=A} \exp\big(\gamma\,\varphi_{q,i}\big), \quad \gamma = \Big[z\sqrt{2 + 4\bar p(1-\bar p) z^2}\Big]_{-\gamma_{\max}}^{\gamma_{\max}},} \tag{9}$$

$$z = \Phi^{-1}\Big(\tfrac{1}{2}\big[1 + \widehat D\,(1 - \nu^2/\zeta^2)_+\big]\Big), \qquad \zeta = \widehat D/\mathrm{SE}, \tag{10}$$

with $\widehat D$ from (5) and φ from (2). At the default $\bar p = \frac{1}{2}$ the exponent is exactly

$$\gamma = z\sqrt{2 + z^2}, \qquad z = \Phi^{-1}(\widehat{\mathrm{AUC}}), \tag{11}$$

one probit and one square root. Nothing in it is fitted to outcomes: ν is a significance level and $\gamma_{\max}$ a clip, both fixed before any data is seen. The clip is not cosmetic, though. Where $\widehat D$ saturates it binds, and the vote then sees $\gamma_{\max}$ rather than the derived magnitude (Section VIII). The dead zone is now visible as a single condition, $|\zeta| \le \nu$, on which γ is identically zero and (9) is bitwise sc by

Proposition 1. Equations (9)–(11) are verified equivalent to the shipped implementation over randomised inputs including every boundary (tests/test_formula.py).

E. Anchor properties

Proposition 1 (Exact sc reduction). At $\gamma = 0$, (3) equals plain sc as a function on every trace pool, including tie-breaks. Under $D = 0$, $P(\gamma = 0) \to 2\Phi(\nu) - 1$ (80% dev, 98% label-free), and γ is continuous through the dead-zone boundary, so a false positive applies an infinitesimal exponent.

Proposition 2 (Exact CISC reduction). With the feature map $\varphi^{\log}_{q,i} = \log c_{q,i} - \overline{\log c_q}$, the weights equal $\kappa_q\, c^{\gamma}_{q,i}$ with a per-item constant $\kappa_q > 0$; hence the argmax, the ties, and the normalized vote shares coincide with CISC-power(γ) on every pool.

Proposition 3 (Regularity). The composite $g(\widehat{D}, \mathrm{SE})$ is continuous, odd, nondecreasing in $\widehat{D}$, nonincreasing in SE in magnitude, with $g(D, 0^+) = \gamma^*(D)$.

Proofs are elementary and pinned by unit tests in the released code (98 tests; the permutation-verified null variance, the EB identity in (7), and the link derivation (8) are each tested numerically).

V. Label-Free Estimation

A. Pipeline

(i) Dedup: single-linkage duplicate groups on the lexical-similarity channel at 0.95; each trace gets weight 1/|group| for plurality determination and pair weighting. (ii) Pseudo-label: $g_{q,i} = \mathbf{1}[a_{q,i} = M_q]$ with M_q the dedup-weighted plurality. (iii) Margin gate: keep the top 60% of items by dedup-weighted margin. (iv) Compute (5) with lab $= g$, giving $(\widehat{D}_g, \mathrm{SE}_g, r_g)$.

B. Sign consistency and its boundary

Proposition 4 (Attenuation identity). Let $\bar{\rho}$ be the pair-weighted probability that an item's plurality is wrong. If the plurality-error event is independent of φ given y (class-conditional noise), then $\mathbb{E}[\widehat{D}_g] = (1 - 2\bar{\rho})\, D$. In particular $\operatorname{sign} \mathbb{E}[\widehat{D}_g] = \operatorname{sign} D$ whenever $\bar{\rho} < 1/2$: the label-free estimate can only under-trust, never mis-sign.

The identity fails when the flip is caused by confidence, that is, under a confident echo. There the observable law under {majority right, $D < 0$} and {majority wrong via confident echo, $D > 0$} is identical (the two-root ambiguity of [10] restated for a single channel), so any label-free guarantee is necessarily conditional; it is stated as such rather than papered over.

C. De-attenuation and alarms

Split-half agreement over R=20 random half-splits estimates $\alpha = p^2 + (1-p)^2/k$ under a one-coin model with k effective wrong alternatives (inverse-Simpson), inverted as $p = [1 + \sqrt{1 - (k{+}1)(1 - k\alpha)}]/(k{+}1)$; $\widehat{D}_g$ is divided by the upper 95% bootstrap bound of $2p - 1$ (floored at 0.2), which can only under-inflate. Four alarms force $\gamma = 0$: duplicate collapse (median Kish ratio < 0.5), sign-aware margin-decoupling, root ambiguity in the split-half quadratic, and insufficient gated items. The margin-decoupling alarm must condition on the estimated trust direction: a sign-naive version ("plurality has the highest mean φ") false-alarms on every benign anti-correlated channel—a defect the author hit, diagnosed, and fixed, and which the released tests pin. Finally the significance gate acts on the raw z (unbiased sign under Proposition 4) and temper on the de-attenuated value. A semi-label-free mode takes only the sign from ∼50 dev labels, routing it into the pipeline and disabling only the proxy-sign alarm; this purchases immunity to the ambiguity above at negligible labeling cost.

VI. Heterogeneity: Impossibility and Escape

A. Per-item adaptation is closed under i.i.d. coupling

Suppose $\kappa_q \overset{\text{iid}}{\sim} \mathcal{N}(0, 0.6^2)$ with no observable covariate.

Proposition 5 (Self-reinforcement). Any per-item rule $\gamma_q = h(\widehat{D}^g_q)$ with h monotone increasing and odd reinforces the plurality on both branches: $\widehat{D}^g_q > 0$ up-weights confident traces, which agree with the plurality; $\widehat{D}^g_q < 0$ up-weights unconfident traces, which are again the plurality side. Empirically such a rule agrees with sc on 97.5% of items and its residual flips are net-harmful (1 right vs. 9 wrong per 400 items).

Proposition 6 (Winner's curse). On plurality-wrong items with $|D_q| > 0.3$, the items where a flip could win, the agreement statistic's sign matches the true sign only 4% of the time.

Proposition 7 (Two-world unidentifiability). For any observed (a, c), the worlds $\{\kappa > 0,\ \text{minority correct}\}$ and $\{\kappa < 0,\ \text{plurality correct}\}$ induce identical observable laws (constructively, D computed against either truth satisfies $D^{w_1} = -D^{w_2}$). No label-free method can separate them.

Consequently the per-item oracle (0.983 in this harness) is unreachable, and the honest behaviour is to fall back to the global estimate, which tact's dead zone does: in the i.i.d. cell every variant returns bitwise sc (zero discordant pairs).

B. TACT-group

Real heterogeneity is typically indexed by an observable covariate (domain, question type). With κ indexed by a group label, running the estimator per group keeps every group inside the operating regime of Sections IV–V; groups with fewer than 30 dev (or 60 unlabeled) items fall back to the global estimate, which Propositions 5–7 show is the only defensible default.

TABLE I
Coupling sweep (accuracy at K=15; 400 paired items per cell; dev n=200). Published protocols sit at the sc floor on the entire negative half-axis.

κ	sc	ECE	devT	SignGrid	tact-dev	tact-LF	oracle
−0.6	.807	.807	.807	1.000	1.000	1.000	1.000
−0.4	.797	.797	.797	1.000	1.000	1.000	1.000
−0.2	.835	.835	.835	.993	.978	.978	.993
−0.1	.762	.762	.762	.892	.880	.885	.892
0.0	.835	.835	.835	.835	.835	.835	.835
+0.1	.795	.795	.917	.917	.902	.902	.917
+0.2	.845	.845	.993	.993	.988	.988	.993
+0.4	.838	.838	1.000	1.000	1.000	1.000	1.000
+0.6	.782	.782	1.000	1.000	1.000	1.000	1.000

VII. Experimental Setup

Harness. A cluster-mixture oracle generates, per item, up to $K_{\max}$=20 cached traces with answers, confidences (1), and two similarity channels; all methods replay identical pools (paired comparisons, exact McNemar tests). Voting budget K=15; 400 items per cell on the sweep, 600 for the group study; dev splits of 200 (primary) and 50 (small-dev).

Regimes. The κ sweep $\{-0.6, \ldots, +0.6\}$; three strictly monotone confidence distortions (compression toward 0.5, over-confident sigmoid, fourth power), rank-preserving by construction, so discrimination is intact while calibration is destroyed; i.i.d. heterogeneity ($\kappa_q \sim \mathcal{N}(0, 0.6^2)$); covariate-structured heterogeneity (three groups at $+0.6/0/-0.6$); and a confident-echo poison (a wrong cluster echoes verbatim with confidence 0.95).

Baselines. sc; CISC-power with $\gamma \in \{0.25, \ldots, 4\}$; CISC-devT, the published dev-calibrated protocol (positive grid picked on dev); a binary ECE gate; SignGrid-dev, the strongest trivial baseline (signed exponent grid picked on dev); and the test-set oracle over signed fixed exponents as the upper envelope. The group study adds the naive self-referential per-item method as a negative control and the per-item link oracle as the ceiling.

Pre-registered falsifiers. F1: tact-dev significantly below the best fixed-γ CISC at κ=+0.6. F2: either variant significantly below sc anywhere on the sweep. F3: the label-free variant fails to beat the ECE gate on sweep average. F4: CISC-devT or SignGrid-dev matches tact-dev everywhere, including the distortion, heterogeneity, and small-dev cells.

VIII. Results

A. Signed recovery, with and without labels

Table I and Fig. 2 give the sweep. Three observations. First, the published protocols never leave the floor on $\kappa < 0$: CISC-devT's grid is positive-only and the ECE gate never opens (dev ECE ranges 0.10–0.80 across the sweep while the signal's discrimination is perfect at the extremes). Second, the label-free variant matches the 200-label variant nearly point-for-point—at κ=−0.6 the raw

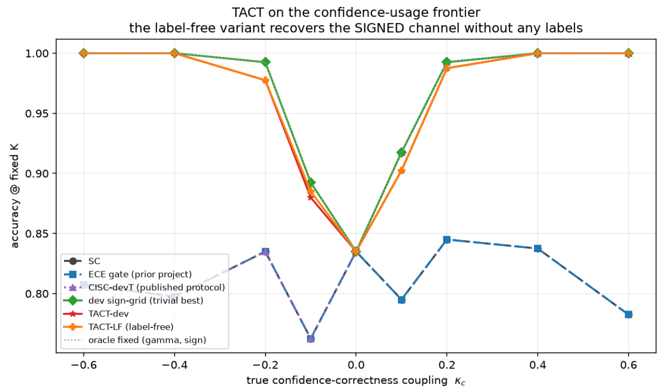

Fig. 2. Main result on the confidence-usage frontier. tact-dev and the fully label-free tact-LF track the signed oracle across the sweep; CISC-devT and the ECE gate sit at the sc floor for all $\kappa < 0$.

TABLE II
Adversarial regimes (accuracy at K=15). "Oracle" is the test-set best over raw-value weight policies; rank invariance beats that entire family under compression.

Regime	sc	devT	SignGrid	tact-dev	tact-LF
Monotone compress	.795	.965	.965	1.000	1.000
Monotone overconf	.795	1.000	1.000	1.000	1.000
Monotone power	.795	1.000	1.000	1.000	1.000
Hetero (i.i.d.)	.810	.810	.810	.810	.810
Confident echo	.200	.200	.550	.585	.200[†]

[†]alarm fires and the method refuses to leave sc—the conditional guarantee of Prop. 4 working as stated.

agreement statistic is $\widehat{D}_g = -0.81$ with $z = -17.6$, and the CCN identity's sign guarantee holds as predicted, yielding 1.000 with zero labels. Third, at $\kappa = 0$ the dead zone returns $\gamma = 0$ exactly, so the paired accuracy difference to sc is identically zero—"non-inferior" is replaced by "identical."

Where the derived exponent actually operates. The four cells that carry the headline 1.000 ($\kappa = \pm 0.4, \pm 0.6$) are cells where $\widehat{D}$ saturates at ± 1, so the link returns $\gamma^* \approx \pm 38$ and the cap $\gamma_{\max}$ is what the vote sees; the derived magnitude is not doing the work there, the sign is. Conversely, at $\kappa = \pm 0.1, \pm 0.2$, where the derived value lands strictly inside the cap (± 1.06 to ± 2.80), tact-dev trails the dev-picked signed grid on all four cells (0.880 vs. 0.892; 0.902 vs. 0.917; 0.978 vs. 0.993; 0.988 vs. 0.993). Read together: against the published protocols the advantage is large and comes from representing the sign at all, whereas against a signed grid the analytic map is not better at choosing a magnitude on these cells. Its advantage over the grid is elsewhere, in the three cells named in C4, and the one place the interpolation itself pays is confident echo, where $\gamma = -1.198$ falls between grid points and beats the grid optimum $\gamma = -1$ (0.585 vs. 0.550).

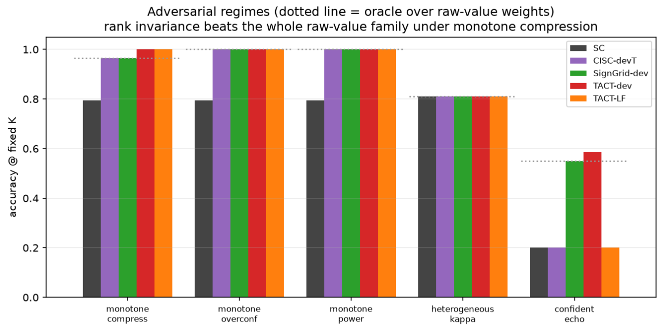

Fig. 3. Adversarial regimes. Dotted line: the oracle over raw-value weights. Left group of bars: rank invariance beats that family under compression; right: the labeled variant counters the confident echo while the label-free variant alarms and refuses.

B. Rank invariance where raw values fail

Under monotone compression (Table II, Fig. 3) all confidences huddle near 0.5, so every c^{γ}-family weight is nearly uniform: even the oracle over raw-value policies reaches only 0.965. tact's rank scores are untouched by the distortion and both variants reach 1.000. Under the confident echo, dev labels reveal the inversion (high confidence $\Rightarrow$ wrong) and tact-dev counters with $\gamma = -1.20$, the best result in the field (0.585; three times the sc floor); label-free, the duplicate-collapse alarm fires and the method correctly refuses—by Proposition 7 no label-free method could do better than a coin flip on the sign here, and pretending otherwise would be the real failure.

C. Heterogeneity

Table III and Fig. 4 give the group study. In the covariate-structured cell, per-group tact recovers each group's signed coupling (dev $\{+4.0, 0.0, -4.0\}$, label-free $\{+2.0, 0.0, -2.0\}$, the κ=0 group correctly dead-zoned—and cracks the floor that provably binds every global policy: the label-free variant reaches 0.940, within 0.007 of the per-item link oracle, with zero paired losses to sc over 600 items ($+79/-0$, $p = 3.3 \times 10^{-24}$). In the i.i.d. cell every legitimate method sits at the floor with zero discordant pairs, and the naive self-referential control lands slightly below it—the empirical face of Propositions 5–7. One observation is reported as-is rather than tuned for: the label-free variant outperforms the dev variant in the grouped cell (0.940 vs. 0.923) because its lower exponent cap (2 vs. 4) regularizes better when $|D| \approx 1$; cap robustness is left as an ablation.

D. Small dev sets and falsifiers

With dev n=50 the conclusions are unchanged (1.000 at $|\kappa|$=0.6; 0.978 at -0.2): the SE-aware shrinkage degrades smoothly rather than catastrophically. All four falsifiers survived: F1 (1.000 vs. 1.000), F2 (bit-identical at κ=0; never significantly below sc elsewhere), F3 (sweep means 0.954 vs. 0.811), and F4 (the distortion and echo cells are unreachable by either grid baseline). Against SignGrid-dev the honest margin is narrow on the homogeneous sweep—tact even trails by 0.005–0.015 in the mid-range, the deliberate cost of shrinkage—and the net advantage concentrates exactly where pre-registered: distortion (+0.035), echo (+0.035), and label-free operation, which no grid can perform.

TABLE III
Heterogeneity study (600 paired items; K=15).

Method	Grouped	i.i.d.
sc (floor)	.808	.827
tact global (dev)	.808	.827
tact-group (dev)	.923	.827
tact-group (label-free)	.940	.827
Naive per-item (neg. control)	.803	.820
Per-item link oracle (ceiling)	.947	.983

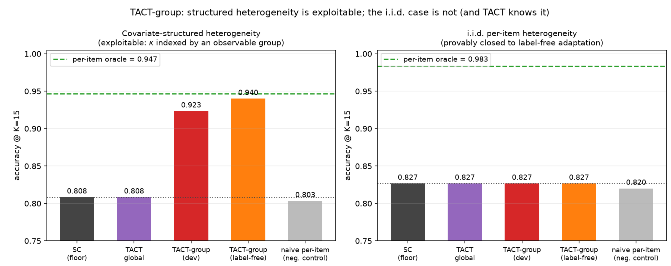

Fig. 4. Structured vs. i.i.d. heterogeneity. Left: with an observable covariate, per-group tact (label-free) approaches the per-item oracle from the 0.808 floor with zero losses to sc. Right: the provably closed i.i.d. cell—every legitimate method at the floor; the negative control slightly below it.

E. Verification of the implementation

Because every claim in Sections IV–VI is a mathematical property rather than an empirical trend, the released code pins each one with an executable test; the suite is 98 tests covering tact and the follow-on work, and runs in 82 seconds. Table IV maps propositions to the tests that would fail if they stopped holding.

Two entries deserve comment. The permutation-invariance test was added after a defect in which the memoisation key made the test pass while the estimator itself was order-dependent by up to 0.10; it now calls the internal routine directly. And the last two rows are counter-tests that assert failure of rejected alternatives — the Kish effective-sample-size formulation and the claim that the shipped default honours the SAFE stopping guarantee — so that neither can be silently reinstated by a later change.

F. Real-trace validation

Validation on real traces used Claude Haiku 4.5 as the frozen model: 100 items (50 GSM8K [23], 50 CommonsenseQA), 12 independent chain-of-thought traces per item with verbalized confidence (1,200 traces total),

TABLE IV
What the test suite verifies. Every proposition in the paper has an executable counterpart; the counter-tests fail deliberately on rejected alternatives so a regression cannot silently reinstate them.

Claim	Test	Evidence
Prop. 1 (exact sc)	gamma_zero_is_ bitwise_sc	200 random pools, identical incl. ties
Dead-zone rate	dead_zone_ probability	>70% under D=0, 300 trials
Prop. 2 (exact CISC)	logval_phi_ reproduces_cisc	identical vote shares, 100 pools
Rank invariance	monotone_ invariance	3 distortions × 100 pools
Null variance	null_variance_ matches_permutation	3,000-draw permutation, 10% tol.
JS–EB identity	js_eb_identity	exact to 10^{-12}
Link (8)	link_values_and_ mixture_correction	closed form, rel. 10^{-9}
Prop. 4 (attenuation)	poisoning_ attenuation_linear	$\rho \in \{.1, .25, .4\}$, abs. .06
Props. 5–7	test_tact_group.py	97.5% sc agreement; 4% sign match
Estimator permutation-invariance	estimator_is_ permutation_invariant	bypasses the memo (regression test)
Rejected: Kish ESS	kish_fails_T2_T3	asserts the failure
Rejected: SAFE guarantee under VoI	frozen_default_ breaks_guarantee	asserts the violation

evaluated at K=12 with a 40/60 dev/test split. Four findings.

(a) The calibration–discrimination distinction reverses on real data, and tact reads it correctly. The channel is extremely well calibrated in the usual sense: ECE = 0.016, far inside the 0.10 gate, so a binary ECE gate opens and hands the channel to CISC. Yet the measured within-item discrimination is $\widehat{D} = -0.219$ with SE = 0.176 ($z = -1.24$): no usable signal, and what little there is points the wrong way (math −0.515, commonsense −0.173; both groups negative). This is the exact mirror image of the synthetic case in which ECE wrongly closed the gate on a discriminative channel (Section III): on real traces ECE wrongly opens it on a non-discriminative one. Calibration is uninformative about voting utility in both directions, and a signed discrimination statistic is what distinguishes them.

(b) The dead zone fires, and costs exactly nothing. With $|z| < \nu$, tact-dev, tact-LF and tact-group all return $\gamma = 0$ and are bit-identical to sc on every test item ($+0/-0$ discordant pairs, $p = 1$). All methods score 0.917. This is the pre-registered null-direction prediction of Section IX confirmed on real data: where the channel carries no signal, the method is free.

(c) Saturation is the binding constraint, not the estimator. Trace-level accuracy is 0.958 on GSM8K and 0.847 on CommonsenseQA, so only 12 of 100 items contain both a correct and an incorrect trace, the only items a within-item rank statistic can use. The estimator is not underpowered by design; the benchmark simply does not present the model with enough genuine uncertainty. Exposing non-null coupling on a strong model requires harder item pools, not more traces per item.

(d) Verbalized confidence is tie-heavy. Two values (0.99, 0.95) account for 49% of all reports, activating the tie-safe degeneration path of (2) on many items.

Scope of this first campaign: one model, two benchmarks, K=12. It confirms the null-direction prediction and the calibration–discrimination argument, and it is not evidence that tact improves accuracy, since the channel carried no signal to exploit. Finding (c) predicts what to do about that, and Section VIII-G does it.

G. Confirmatory campaign on harder items

Finding (c) predicts that a channel measured as null on saturated benchmarks should become measurable on items the model finds genuinely uncertain. A pre-registered follow-up tests that prediction: 119 MATH level-5 problems [24], [25], 16 traces each from the same frozen model, a 30-item sign set and an 89-item evaluation set drawn from the registered list before any trace was collected, and five hypotheses (H1–H5) fixed in advance.

The channel is real. On the evaluation set the pooled statistic is $\widehat{D} = +0.250$ with SE = 0.098, so $z = +2.54$ and H1 passes. This is the first real-trace evidence that verbalized confidence carries positive within-item discrimination; the same measurement on GSM8K/CommonsenseQA gave −0.219 ($z = -1.24$).

The endpoint was unpassable for any method. The realized substrate saturated again: per-trace accuracy 0.819, sc 0.888, a decisive stratum of 10 of 89 items, and the correct answer present in the pool on only 4 of those. The in-pool oracle therefore tops out at $+4/-0$, exact one-sided $p = 0.0625$. H2 fails, but it fails for every conceivable aggregation method including a perfect one, so the failure is a property of the substrate rather than of the estimator.

Abstention behaved as designed. tact returned $\gamma = 0$, with alarms E4 and E2 firing on the label-free path and the sign set holding too few informative items to supply a semi-label-free sign. The vote is therefore bit-identical to sc at 0.888 (H3, H4 pass). The cost of acting anyway is visible in the same table: best-single-confidence, the trivial baseline that always trusts the channel, loses 4.5 points at 0.843.

One caveat from this campaign transfers beyond tact. Measured difficulty depended on the collection protocol: a 30-problem-per-call probe put level-5 plurality accuracy at 0.40, while the 15-problem-per-call confirmatory run yielded 0.888 on the same stratum. Batch size belongs in the experimental record whenever traces are collected in batches.

H. How wide is the addressable stratum?

Both campaigns failed their endpoint for the same reason, which suggests measuring that reason directly.

TABLE V
Confirmatory campaign, MATH level-5 evaluation set (89 items, K=16). Every method replays the same cached pools. The duplication channel is inert because no reasoning text was collected, so dedup-sc coincides with sc by construction.

Method	Accuracy	net vs. sc
sc / dedup-sc / CISC-linear	0.888	—
tact-LF	0.888	$+0/-0$
tact-semi-LF	0.888	$+0/-0$
best-single-confidence	0.843	$+1/-5$
in-pool oracle (ceiling)	0.933	$+4/-0$

TABLE VI
The addressable stratum across substrates. Rows marked † are measured here; HumanEval+/MBPP+ is recomputed from published oracle-minus- selector tables. As items harden they pass from saturated to capability-limited without the window widening.

Domain	Substrate	Window
QA, label-free	GSM8K / CommonsenseQA†	12% informative, 9% decisive
QA, label-free	MATH level-5†	11% decisive, 4% rescuable
QA, label-free	AIME / AMC†	23% decisive, 3% rescuable
Code, executable	HumanEval+ / MBPP+	3.56%
Code, executable	LeetCode Med/Hard†	7.5%

Define the window as the fraction of items where the plurality is wrong and the correct answer is present in the pool: the ceiling for any label-free aggregation method, since nothing outside it can be changed.

The window was measured on five substrates spanning two domains (Table VI). For code generation, where an executable test suite supplies per-sample ground truth and the window might reasonably be expected to widen, 40 LeetCode Medium/Hard problems [26] were solved 8 times each and graded against the benchmark's hidden suites, with the baseline taken as the largest behavioural cluster over probe inputs (never expected outputs). The window is $3/40 = 7.5\%$ (CI_{95} 2.6–19.9%): wider than label-free QA, but the same order, and the composition is the same shape at 30 saturated, 7 capability wall and 3 rescuable, i.e. 75/17.5/7.5%. Nor does budget open it. The seven capability-wall problems produced zero correct solutions in 224 further attempts (per-problem 95% upper bound on the pass rate 0.088), and extrapolating oracle@N shows the window saturating by N=32.

One precaution belongs with these numbers, because omitting it would have inverted them. The grading harness was validated against the benchmark's own reference solutions before any candidate was scored: 178 of 180 pass under the sandbox's resource limits. An earlier version of the same harness failed 100% of executions because the host rejects one of the requested limits outright, and that condition presents as a candidate failure rather than as an error. Studies that grade by execution should report their reference-solution pass rate for the same reason a calibration curve is reported: without it, a broken harness and a capability wall look identical.

IX. Discussion and Limitations

What the evidence does and does not show. The accuracy claims are all on a synthetic oracle whose confidence model (1) is, at the homogeneous cells, the very coupling the estimator measures. Three design choices limit the circularity: the adversarial regimes (distortions, heterogeneity, echo) lie outside the estimator's working model; mechanism-recovery claims (does $\widehat{D}$ track κ?) are reported separately from accuracy claims; and the pre-measured baseline landscape (Fig. 1) fixed the winnable cells before the method existed. The real-trace campaigns of Sections VIII-F and VIII-G test the premise and the abstention behaviour, and both predictions held: the channel is null on saturated benchmarks and positive on competition mathematics, and the dead zone kept the vote bit-identical to sc in each case. They do not test the accuracy claim, because on neither substrate was the addressable stratum large enough for any method to demonstrate a gain (Section VIII-H).

Narrow margins where labels abound. When labels are plentiful and the confidence scale is trusted, a dev-picked signed grid captures most of the value; tact's case rests on the label-free setting, distorted scales, small dev sets, and the exactness of its anchors.

Conditional label-free guarantee, and what happens past the boundary. Proposition 4 requires $\bar{\rho} < 1/2$ after deduplication, and the confident-echo ambiguity is fundamental (Proposition 7). Follow-on work measured the consequence of crossing that boundary, and it is worse than under-trust. In a paraphrased wrong-majority cell (a dominant wrong cluster that is semantically tight but carries no verbatim signature, so deduplication has nothing to collapse) the plurality is wrong on most items, $\bar{\rho} > 1/2$, and tact-LF does not merely shrink toward sc: it mis-signs, saturates at $\gamma = -2.0$, and scores 0.000 against an sc floor of 0.340. None of the four alarms fires, because E1 keys on verbatim duplication which is absent by construction. This is the method's sharpest unguarded failure mode: the guarantee is conditional, the condition is not observable label-free, and the existing diagnostics do not detect its violation. Where a systematically wrong majority is plausible, the semi-label-free mode (sign from $\sim$50 labels) should be the default rather than an optional refinement.

Global exponent per group. Within a group, tact ships one exponent; per-item variation inside a group is unexploitable by Propositions 5–7 unless further covariates exist.

The thin window. Section VIII-H measures the stratum this whole family of methods can act on at 3–7.5% of items on every substrate tried, in two domains, with no widening as items harden: they pass from saturated straight to capability-limited. Two consequences follow for the method proposed here. First, abstention is not a conservative compromise but the only correct default, and the measured cost of acting anyway was negative on both real substrates (best-single-confidence loses 4.5 points in Table V where the dead zone holds tact at the sc floor). Second, an aggregation gain of the size reported on the synthetic harness is not measurable on a benchmark of a few hundred items at these window widths, which is why the real-trace claim in this paper is confined to the premise (the channel exists and is signed) and to the abstention behaviour, and does not extend to accuracy. Demonstrating the gain needs a (model, benchmark) pair whose plurality is wrong on 30–60% of items with the correct answer still reachable, and no pair tried here satisfies both.

X. Conclusion

tact turns "how much should this model's confidence be trusted?" into a measured, signed, uncertainty-aware quantity with an exact fallback at one end, plain self-consistency whenever the evidence is absent, and an exact correspondence to CISC-power at the other under the log-value feature map (the shipped default uses van der Waerden scores, so that correspondence is a property of the family, not of the shipped configuration), and shows that the sign, long unrepresentable in this family of methods, can be recovered without any labels under stated and tested conditions. The accompanying impossibility results draw the boundary that any future per-item method must respect, and the falsification protocol, having already killed one of the author's own systems, is offered as the more portable contribution.

References

[1] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," in Proc. ICLR, 2023. https://arxiv.org/pdf/2203.11171

[2] K. Tian et al., "Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback," in Proc. EMNLP, 2023. https://aclanthology.org/2023.emnlp-main.330.pdf

[3] M. Xiong et al., "Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs," in Proc. ICLR, 2024. https://openreview.net/pdf?id=gjeQKFxFpZ

[4] S. Kadavath et al., "Language models (mostly) know what they know," arXiv:2207.05221, 2022. https://arxiv.org/pdf/2207.05221

[5] A. Taubenfeld et al., "Confidence improves self-consistency in LLMs," in Findings of ACL, 2025, arXiv:2502.06233. https://aclanthology.org/2025.findings-acl.1030.pdf

[6] J. Kim, N. Yang, K. Min, and K. Jung, "Reliability-aware adaptive self-consistency for efficient sampling in LLM reasoning," in Findings of ACL, 2026, pp. 21575–21590. https://aclanthology.org/2026.findings-acl.1085.pdf

[7] Y. Fu et al., "Deep think with confidence," arXiv:2508.15260, 2025. https://arxiv.org/pdf/2508.15260

[8] X. Huang, S. Li, M. Yu, M. Sesia, H. Hassani, I. Lee, O. Bastani, and E. Dobriban, "Uncertainty in language models: Assessment through rank-calibration," in Proc. EMNLP, 2024, pp. 284–312. https://aclanthology.org/2024.emnlp-main.18.pdf

[9] A. P. Dawid and A. M. Skene, "Maximum likelihood estimation of observer error-rates using the EM algorithm," J. Roy. Statist. Soc. C, vol. 28, no. 1, pp. 20–28, 1979. https://doi.org/10.2307/2346806

[10] F. Parisi, F. Strino, B. Nadler, and Y. Kluger, "Ranking and combining multiple predictors without labeled data," Proc. Natl. Acad. Sci., vol. 111, no. 4, pp. 1253–1258, 2014. https://pmc.ncbi.nlm.nih.gov/articles/PMC3910607/pdf/pnas.201219097.pdf

[11] Y. Li et al., "Making language models better reasoners with step-aware verifier," in Proc. ACL, 2023. https://aclanthology.org/2023.acl-long.291.pdf

[12] Z. Kang, X. Zhao, and D. Song, "Scalable best-of-N selection for large language models via self-certainty," in Proc. NeurIPS, 2025, arXiv:2502.18581. https://proceedings.neurips.cc/paper_files/paper/2025/file/1c7eff166a8e345f664f0faa8f4e4d2e-Paper-Conference.pdf

[13] P. Aggarwal, A. Madaan, Y. Yang, and Mausam, "Let's sample step by step: Adaptive-consistency for efficient reasoning and coding with LLMs," in Proc. EMNLP, 2023, pp. 12375–12396. https://aclanthology.org/2023.emnlp-main.761.pdf

[14] Y. Li et al., "Escape sky-high cost: Early-stopping self-consistency for multi-step reasoning," in Proc. ICLR, 2024. https://arxiv.org/pdf/2401.10480

[15] J. Whitehill et al., "Whose vote should count more: Optimal integration of labels from labelers of unknown expertise," in Proc. NeurIPS, 2009. https://proceedings.neurips.cc/paper_files/paper/2009/file/f899139df5e1059396431415e770c6dd-Paper.pdf

[16] D. R. Karger, S. Oh, and D. Shah, "Iterative learning for reliable crowdsourcing systems," in Proc. NeurIPS, 2011. https://proceedings.neurips.cc/paper_files/paper/2011/file/c667d53acd899a97a85de0c201ba99be-Paper.pdf

[17] J. Lee, V. Ma, S. Zhao, Y. Nair, A. Spector, R. Cohen, and E. J. Candès, "FUSE: Ensembling verifiers with zero labeled data," arXiv:2604.18547, 2026. https://arxiv.org/pdf/2604.18547

[18] R. Ai, Y. Pan, D. Simchi-Levi, M. Tambe, and H. Xu, "Beyond majority voting: LLM aggregation by leveraging higher-order information," arXiv:2510.01499, 2025, accepted to ICML 2026. https://arxiv.org/pdf/2510.01499

[19] P. van Elteren, "On the combination of independent two-sample tests of Wilcoxon," Bull. Int. Statist. Inst., vol. 37, pp. 351–361, 1960. https://catalog.hathitrust.org/Record/008896012

[20] W. James and C. Stein, "Estimation with quadratic loss," in Proc. 4th Berkeley Symp. Math. Statist. Prob., 1961, pp. 361–379. https://digitalassets.lib.berkeley.edu/math/ucb/text/math_s4_v1_article-19.pdf

[21] L. Kish, Survey Sampling. New York, NY, USA: Wiley, 1965. https://www.wiley.com/en-us/Survey+Sampling-p-9780471109495

[22] J. N. K. Rao and A. J. Scott, "The analysis of categorical data from complex sample surveys," J. Amer. Statist. Assoc., vol. 76, no. 374, pp. 221–230, 1981. https://doi.org/10.1080/01621459.1981.10477633

[23] K. Cobbe et al., "Training verifiers to solve math word problems," arXiv:2110.14168, 2021.

[24] D. Hendrycks et al., "Measuring mathematical problem solving with the MATH dataset," in Proc. NeurIPS Datasets and Benchmarks, 2021. https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/file/be83ab3ecd0db773eb2dc1b0a17836a1-Paper-round2.pdf

[25] H. Lightman et al., "Let's verify step by step," in Proc. ICLR, 2024. (MATH-500 test subset.)

[26] Y. Xia et al., "LeetCodeDataset: A temporal dataset for robust evaluation and efficient training of code LLMs," arXiv:2504.14655, 2025. https://arxiv.org/pdf/2504.14655